

An Empirical Evaluation of Practical Machine Learning for Surface Electromyography-Based Prosthetic Hand Control

Joseph Letobar^{1,*}

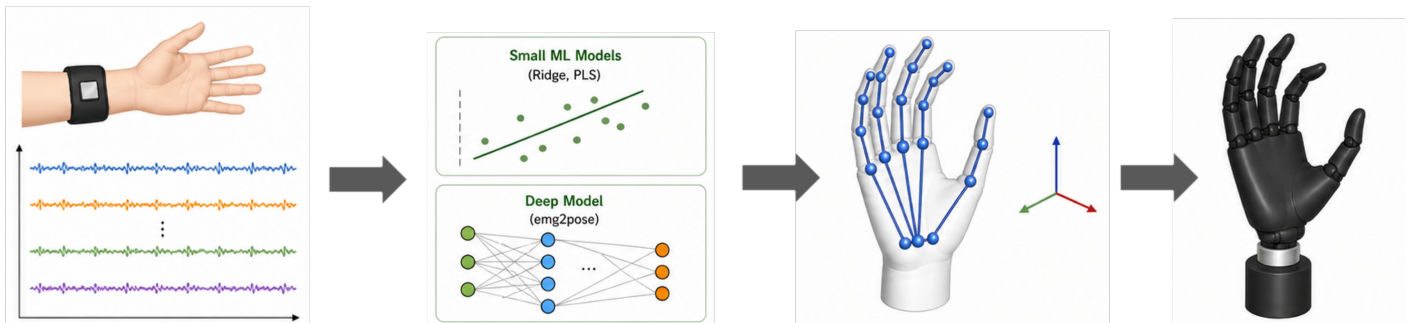
¹Department of Computer Science, University of Central Florida

* Corresponding author. joseph.letobar@ucf.edu

Abstract

Surface electromyography (sEMG) captures electrical activity from muscles through the skin, and when measured at the wrist, machine learning models can infer the hand gesture or pose that produced this signal. Meta Reality Labs provide the largest high-quality dataset of sEMG signals paired with labeled hand poses, enabling large-scale empirical analysis. Their accompanying model performs well under generalization across people due to scale in both data and model capacity, similar to trends seen in large language models. This work addresses a different question from generalization, focusing instead on user personalization within prosthetic hand control, where predicted gestures from EMG signals are used as commands for the prosthetic hand to execute. The focus is on user specificity, low latency, and limited data settings to provide practical and targeted insights for future work in this space.

Graphical Abstract



Introduction

Electromyography (EMG) has been used for decades to control prosthetic hands, referred to as myoelectric control (1). More recently, machine learning methods have been shown to improve control performance and intuitiveness (3). In these systems, EMG signals are measured using electrodes (sensors) placed on the skin and machine learning models map them to gestures for the prosthetic hand to execute. This work focuses on non-invasive surface electromyography (sEMG), which can be measured directly from the skin without surgical intervention, but introduces practical challenges such as sensitivity to electrode placement, sweat, and muscle fatigue (2; 5).

Despite significant research effort, commercially used myoelectric prosthetic hands continue to exhibit high abandonment rates of approximately 40% (4).

One reason for this high abandonment rate is the lack of simultaneous control of multiple degrees of freedom (DoFs), for example continuously controlling both index finger flexion and thumb abduction at the same time. Most existing systems simplify the learning problem to discrete gesture classification, such as open or close hand, limiting the expressiveness and intuitiveness of control (5).

More precise decoding approaches that regress individual joint positions or angles have shown promising results. Recent work from Meta Reality Labs, specifically the emg2pose model, demonstrates strong performance and is used as a baseline in this work (6). However, these approaches rely on deep sequence-based neural architectures and rely on large-scale data and model capacity, making them less practical for the embedded systems found on prosthetic hands where low latency and power consumption are critical (8; 9; 7).

Their model also focuses on generalization across users, targeting the application of more general human-computer interaction, where a single model can be used by many individuals without retraining, as seen in Meta’s EMG-based input systems for augmented reality glasses (8).

This work performs an empirical analysis of the large-scale dataset provided by Meta Reality Labs. This dataset is the largest published to date, consisting of recordings from 193 users, each user with sessions amounting to approximately 30 minutes of data, corresponding to different hand movement sequences, with synchronized ground truth joint angles obtained via motion capture (6).

This study operates under constraints relevant to prosthetic deployment, including user personalization, low latency, and limited computational cost (5; 7). A single user is selected, and models are trained on their sessions and evaluated on a held-out session. Performance is assessed in terms of accuracy, smoothness, and latency.

Results show that, under these constraints, small classical machine learning models can achieve accuracy that is competitive with large-scale deep models while operating with orders of magnitude lower latency. However, these models may exhibit reduced robustness to variability in EMG signals, including changes that occur even within a single user over time. Additionally, these models produce less smooth predictions, which can be partially mitigated through post-processing. While the large model performs best overall, this work provides practical insights into when lightweight regression models may be preferable, depending on factors such as available data, hardware constraints, and intended use of the prosthetic system.

Material and Methods

Dataset

This work uses the emg2pose benchmark dataset from Meta Reality Labs (6). It contains high-fidelity wrist sEMG recordings paired with hand pose labels. The sEMG signals are recorded using the Reality Labs sEMG-RD wristband at 2 kHz, and ground truth joint angles are obtained from a 26-camera motion capture system.

The dataset includes 193 unique users performing 29 variations of hand gestures, referred to as “stages” (6). Each user has at least 30 sessions, with each session lasting approximately one minute. In total, the dataset comprises roughly 370 hours of recorded data.

Experimental Setup

From the 193 users in the dataset, one user is randomly selected. All of that user’s sessions are used for training across the evaluated machine learning models, while one session is randomly held out for evaluation to prevent data leakage.

This process is repeated for 40 iterations, where each iteration consists of selecting a user, training all models on their sessions, evaluating on a held-out session, and computing metrics. Repeating this process accounts for variability across users, as some users may exhibit more challenging EMG signals due to factors such as electrode placement, fatigue, and individual physiological variability, or perform more difficult movement sequences. (5; 10)

All iterations are aggregated and averaged to obtain a representative set of performance metrics across the dataset.

Meta emg2pose Baseline

The Meta emg2pose model is designed for user-generalized EMG-to-pose regression and is evaluated here using the pretrained checkpoint provided by Meta, which was trained on all users (minus held out).

The pretrained emg2pose model is used as a reference to the state of the art rather than as a fully symmetric comparison. Training this architecture from scratch under the experimental data constraints was also attempted, but it failed to converge to meaningful performance and was therefore excluded from further analysis, indicating the need to study lightweight machine learning methods in this setting.

The model predicts joint angular velocities, which are integrated over time to obtain joint angle estimates (6). This is done because EMG signals are more directly related to muscle activation and movement than absolute joint positions (11).

Raw sEMG signals sampled at 2 kHz are passed through a convolutional front-end that performs feature extraction and temporal downsampling, reducing the effective sampling rate to approximately 50 Hz. At this rate, each time step corresponds to roughly 20 ms of signal. These features are then processed by a temporal convolutional (TDS) network to produce a learned representation at each time step. The resulting features are concatenated with the previous joint angle estimate and passed into an LSTM decoder, which predicts joint angular velocities for the next time step.

The model is autoregressive because each prediction depends on the previous joint angle estimate, which is fed back into the decoder at the next time step. The predicted velocities are then integrated over time to produce joint angle estimates. The pretrained model checkpoint provided by Meta Reality Labs is used without modification, and no additional training or fine-tuning is performed for the final evaluation.

ConvLSTM Model

The ConvLSTM model is a lightweight, autoregressive sequence model for EMG-to-pose regression. These design choices are broadly consistent with prior work in EMG-based hand pose estimation, which leverages temporal models such as LSTMs and convolutional feature extraction for decoding muscle activity into joint motion (12; 9). Input sEMG signals are first temporally downsampled by a factor of 2, reducing the effective sampling rate and computational cost. The signal is then segmented into fixed-length sequences of 100 time steps, where each sequence is used to predict the joint angles at the final time step.

At each time step, the input consists of the sEMG signal concatenated with the previous joint angle estimate, resulting in an input dimensionality of $C + D$, where C is the number of EMG channels and D is the number of joint outputs. The model is autoregressive, as predictions depend on previous outputs.

The input sequence is passed through a convolutional front-end consisting of two 1D convolutional layers, which expand the signal to 32 channels and then compress it to 16 channels while extracting local temporal features. These features are then processed by a single-layer LSTM with a hidden size of 64, which models temporal dependencies across the sequence.

The LSTM output is passed through a linear layer to produce the final joint angle prediction. This architecture is intentionally compact, prioritizing low latency and computational efficiency while retaining the ability to model temporal structure in the EMG signal.

Classical Machine Learning Models (Ridge, PLS)

Classical ML models operate on engineered features extracted from raw sEMG signals rather than the raw signal itself. Following standard practice in the EMG literature (13), a set of time-domain features is computed over sliding windows, including metrics such as mean absolute value, waveform length, variance, and zero-crossings. These features are computed over fixed-length windows with a defined stride, introducing overlap between consecutive samples and thereby incorporating short-term temporal structure into the representation.

The windowing and stride are chosen such that the resulting feature sequence is effectively downsampled to an effective rate of approximately 30 Hz, which is more representative of real-time prosthetic control constraints (5). As a result, each feature vector summarizes a short segment of EMG activity, capturing both amplitude and frequency-related characteristics in a nonlinear way (13). Removing this overlapping, windowed structure resulted in a noticeable degradation in performance.

Ridge regression and partial least squares (PLS) regression are then applied to map these features to joint kinematics. Although both models are linear in form, they operate on a feature space that is already a nonlinear transformation of the raw EMG signal. This allows them to approximate nonlinear relationships between muscle activation and movement without requiring explicitly nonlinear models (11).

Ridge regression learns a direct linear mapping from the engineered feature space to the output, relying entirely on the feature representation to capture relevant structure. In contrast, PLS first projects the feature space into a lower-dimensional latent space that maximizes covariance with the target outputs, allowing it to capture interactions and redundancy among features before performing regression.

To further incorporate temporal dependencies, an autoregressive structure is introduced through lagged outputs. Specifically, previous model predictions are concatenated with the current feature vector and used as input for the next prediction step. Removing these lagged inputs also degraded performance, indicating that sequential dependency plays an important role in stabilizing predictions over time.

Evaluation Metrics

All models were evaluated along three dimensions: accuracy, smoothness, and latency. Accuracy was measured using two complementary metrics.

For consistency with the original benchmark, all evaluation code except latency was adopted directly from Meta's implementation (6).

Additionally, when ground truth joint angles were unavailable or invalid due to motion capture failures, those time steps were excluded from all metrics to avoid corrupting the results.

Angular Error

Angular error measures the difference between predicted and ground truth joint angles. This is computed as the mean absolute error (MAE) over all joint angles.

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{\theta}_i - \theta_i|$$

where $\hat{\theta}_i$ and θ_i denote the predicted and ground truth joint angles, respectively. The error is reported in degrees.

Landmark Distance

Landmark distance evaluates the spatial accuracy of the reconstructed hand. Predicted joint angles are passed through a *hand model* provided by Meta which obtains joint positions in 3D space (6).

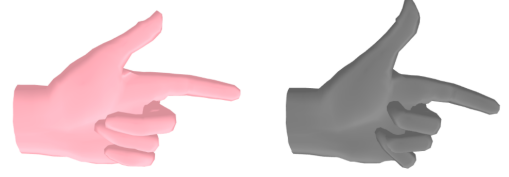


Figure 1 Visualization of the hand model used to compute landmark distance, showing ground-truth (left, pink) and predicted (right, gray) hand poses illustrating spatial differences.

The error is then computed as the Euclidean distance between predicted and ground truth joint positions, averaged across all joints.

$$Error = \frac{1}{N} \sum_{i=1}^N |\hat{\mathbf{p}}_i - \mathbf{p}_i|$$

where $\hat{\mathbf{p}}_i$ and $\mathbf{p}_i$ are the predicted and ground truth 3D positions of joint i . The error is reported in millimeters.

Jerk

Jerk measures the smoothness of the predicted motion. It is computed by taking successive discrete differences of the predicted joint angles and averaging the absolute value, approximating the third derivative with respect to time:

$$jerk = \frac{1}{N} \sum_{i=1}^N \left| \frac{d^3 \theta_i}{dt^3} \right|$$

where θ_i denotes the joint angle at time step i . Lower values indicate smoother motion.

Latency

Latency was evaluated under a simulated streaming setting. Raw EMG samples were processed sequentially, with a new prediction made every 50 samples (stride) using the most recent 100 samples (window). This results in overlapping windows, where each new inference reuses the last 50 samples from the previous window. This setup approximates real prosthetic deployment (5).

Since latency is not reported in the original Meta benchmark, this metric is introduced here as a new evaluation.

Latency was measured as the elapsed time between invoking the model and receiving its prediction using Python's high-resolution timer. The median latency across all inferences was reported:

$$\mathcal{T} = \{t_2^{(i)} - t_1^{(i)}\}_{i=1}^N$$

$$\text{latency} = \text{median}(\mathcal{T})$$

where $t_1^{(i)}$ and $t_2^{(i)}$ denote the start and end times of inference i , respectively, and $\mathcal{T}$ contains latencies for all inferences.

Results

All models were evaluated under the experimental setup described previously. Each trial was run multiple times, and results were averaged to obtain a representative estimate of performance across the data. Metrics are reported under the *seen-user setting*, where each model is evaluated on a held-out session from the same user used during training.

Exponential moving average (EMA) smoothing was applied as a post-processing step to reduce noise in model predictions. Its effect on smoothness is reported as the change in jerk relative to the unsmoothed output. None of the changes in other metrics due to EMA smoothing were significant.

Table 1 Seen-user performance with EMA smoothing effects on jerk and latency. AE = Angular Error, LD = Landmark Distance.

Model	AE (°)	LD (mm)	Jerk (°)	Δ Jerk (EMA)	Latency (ms)
ConvLSTM	16.78	24.34	5675.87	-4269.69	0.68
emg2pose	13.38	19.81	0.75		73.99
PLS	16.81	23.25	1260.25	-1093.81	0.22
Ridge	15.25	21.52	381.25	-297.68	0.23

Each model is reported across the evaluation metrics shown in Table 1. As expected, the emg2pose model achieved the lowest error due to its scale and training data. The metric was also close to Meta's reported, showing effective replication. However, the primary question is whether smaller models can perform competitively under user-specific training conditions. The results indicate that they can, with error differences compared to emg2pose on the order of approximately $\sim 2-4^\circ$ in angular error and $\sim 2-5$ mm in landmark distance.

The most significant differences arise in latency and smoothness. The smaller models achieve substantially lower latency, with up to $300\times$ lower latency than emg2pose in the best case, which is favorable for real-time prosthetic control. However, this comes at the cost of increased jerk, indicating less smooth predictions. Applying EMA smoothing reduced jerk by an average of 80% across all models, bringing the outputs to a usable level for prosthetic control. As mentioned, changes in other metrics were not significant. The implications of this trade-off are discussed further in the Discussion.

Among the evaluated models, ridge regression achieved the best overall balance across metrics, outperforming other classical ML methods and even the small neural model (ConvLSTM), which underperformed relative to classical ML approaches, particularly in the jerk metric.

Additionally, performance was evaluated on unseen users to the model to assess generalization. As expected, performance degraded substantially for the smaller models, while the emg2pose model maintained its performance. Since this is not the primary focus of the study, a more detailed analysis is deferred to the Discussion.

Table 2 Unseen-user performance with EMA smoothing effects on jerk and latency. AE = Angular Error, LD = Landmark Distance.

Model	AE (°)	LD (mm)	Jerk (°)	Δ Jerk (EMA)	Latency (ms)
ConvLSTM	20.20	28.42	4617.23	-3678.55	0.68
emg2pose	13.61	15.38	0.80		73.99
PLS	24.48	30.29	1194.98	-1038.58	0.22
Ridge	32.10	33.12	426.94	-350.23	0.23

Though it is notable that the ConvLSTM model retained relatively stable performance under unseen-user evaluation compared to the classical models. While this result is based on cross-user generalization, it may suggest that sequence-based neural architectures may also be more robust to variability seen within-users, such as changes in electrode placement, muscle fatigue, and session-to-session signal drift (5).

Discussion

Model Performance

Under the seen-user setting, ridge regression emerged as the strongest performer. Despite being a linear model, and EMG signals being inherently nonlinear (11), the use of established EMG feature engineering methods (13), combined with windowed feature extraction, enabled it to capture sufficient structure in the signal. As a result, ridge regression achieved performance that approached that of the emg2pose model in this setting.

PLS performed slightly worse, which may be attributed to its use of dimensionality reduction. Since the input features were already intentionally engineered to capture relevant signal characteristics, this additional compression may have been redundant and removed useful information.

The ConvLSTM model underperformed relative to expectations. This may be due to a mismatch between model capacity and available data. Although relatively small, it remains a neural model, and neural models require sufficient data to learn meaningful representations, which was not the focus of this evaluation.

Notably, the ConvLSTM model also exhibited significantly higher jerk compared to other models. This is unexpected, as recurrent architectures are typically better suited for producing temporally smooth outputs.

To address the underperformance in accuracy and jerk, during implementation, parts of the architecture were modified, including removing the convolutional layer responsible for learned feature extraction. Specifically:

- In one configuration, the convolutional layer was replaced with manually engineered features
- In another, it was removed entirely, allowing the LSTM to operate directly on raw EMG signals

Notably, recurrent models such as LSTMs are commonly used in EMG-based modeling due to their ability to capture temporal dependencies (12; 9). The observed underperformance in this setting is therefore somewhat unexpected, but is likely explained by the limited data regime, as these models typically benefit from substantially larger training datasets.

Prior work has shown that linear regression-based approaches can achieve strong performance in EMG-based control (15; 16). These methods are effective because, although raw EMG signals are complex and nonlinear, engineered feature representations transform the signal into a space where the relationship to muscle activation patterns is approximately linear. This allows relatively simple models to perform well when trained on user-specific data (13).

In contrast, more recent work has explored nonlinear models, including neural networks, in part to improve robustness to real-world variability such as electrode displacement, muscle fatigue, and session-to-session changes (5). Importantly, the dataset used in this

work includes natural variability across users and sessions, and strong performance was still observed under these conditions. This suggests that linear models remain a viable option in practice, particularly in user-specific settings where data is limited and computational constraints are important.

However, it is important to note that the smaller models degraded under cross-user evaluation, with the largest drop observed in ridge regression, followed by PLS, while the ConvLSTM showed the smallest degradation. Although this study is not intended to evaluate generalization, this still provides some insight into robustness. In practice, EMG signals vary not only across users but also within a single user over time due to factors such as electrode placement, muscle fatigue, and session-to-session drift (5). The observed behavior suggests that nonlinear models may better handle this variability, while linear methods remain effective in user-specific settings where conditions are more consistent.

As mentioned earlier, training the emg2pose architecture from scratch within the defined experimental setup (single-user training with held-out session evaluation) was also attempted. In this setting, the model failed to achieve meaningful performance, significantly underperforming all reported methods. This further supports the conclusion that large neural architectures require large-scale data to learn useful representations, and are not effective in low-data, user-specific regimes without pretraining.

Latency–Smoothness Tradeoff and EMA

Among the evaluated models, ridge regression emerged as the strongest overall performer in this setting. However, despite its accuracy, it exhibited relatively high jerk, indicating low temporal smoothness. To address this, an exponential moving average (EMA) was applied as a simple post-processing step to demonstrate how smoothing can improve control stability.

EMA significantly improved smoothness by about 80% on average, but introduces a tradeoff. Because each prediction is partially influenced by previous outputs, this creates a form of temporal lag. This effect can be tuned through the smoothing parameter, but cannot be entirely eliminated. More advanced approaches, such as Kalman filtering, may provide a better balance between responsiveness and stability in future work.

Additionally, the outputs of ridge regression can be qualitatively described as a *simplification* of hand motion. Applying smoothing further amplifies this effect. A qualitative example is shown below, illustrating behavior that is not fully captured by aggregate error metrics.



Figure 2 Qualitative comparison showing ridge regression with EMA smoothing (right, gray) produces simplified, less articulated hand motion relative to ground truth (left, pink).

While not part of the primary evaluation metrics, this qualitative difference can be examined more directly by analyzing per-joint

errors rather than aggregate averages. Per-joint error variance was computed across runs to get a good representation over the dataset.

Ridge regression shows substantially lower variance ($\approx 24 \text{ deg}^2$) in per-joint errors compared to emg2pose ($\approx 69 \text{ deg}^2$), indicating more uniform error distribution across joints. This supports the qualitative observation that ridge produces more correlated, simplified motion patterns, whereas emg2pose attempts higher-fidelity articulation with greater variability across individual joints.

This likely reflects the nature of the optimization objective, where ridge regression minimizes a global squared error and therefore tends to distribute error more evenly across outputs, effectively producing a more averaged solution. In contrast, emg2pose appears to allocate error unevenly across joints, prioritizing higher-fidelity reconstruction of certain degrees of freedom at the expense of others. Future work should explore modifying the optimization objective to explicitly minimize per-joint error, for example by weighting or separating losses across individual fingers, to evaluate whether this improves articulation or degrades overall performance.

This comparison highlights a key tradeoff not fully captured by standard error metrics. Ridge regression with smoothing produces visibly less precise articulation across individual degrees of freedom, effectively simplifying motions into more generalized gesture-like patterns. While this reduces fine-grained joint accuracy, it introduces a level of stability and consistency that can be advantageous in underactuated or low-DoF prosthetic systems, where independent control of individual fingers is not available due to mechanical design.

At the same time, unlike purely discrete gesture classifiers, this approach retains continuous control, allowing for more responsive and adaptive hand behavior. This provides a practical middle ground, where control is smoother and more stable than high-variance models, while still enabling proportional, real-time interaction aligned with the physical constraints of the hardware.

In contrast, fully actuated high-DoF prosthetic systems, which are less common due to mechanical and control constraints (14), larger models such as emg2pose may be more appropriate, provided sufficient computational resources and training data are available.

Additional Observations

As mentioned in Results, all models were also evaluated on unseen users to assess generalization. The pretrained emg2pose model maintained its performance, while all other models degraded significantly, with error increasing by approximately $\approx 3\times$. This is expected, as the classical ML models rely heavily on user-specific feature distributions and do not generalize well without retraining.

Support Vector Regression (SVR) was also explored and, in some cases, achieved performance that slightly exceeded the emg2pose model. However, due to its quadratic computational complexity, it did not scale to full training sessions and was limited to a small subset of the data (approximately 1/30). This makes it impractical for real-world use, where models must be trained or adapted quickly for a new user.

Fine-tuning the pretrained emg2pose model on a held-out user (not seen during Meta’s original training) did not produce any meaningful improvement over the base model. Across repeated runs, fine-tuning changed angular error by approximately $\pm 0.2^\circ$, indicating no consistent or practically significant gain. This suggests that the pretrained model already captures much of the underlying signal structure for this setting.

Future work should evaluate fine-tuning under different data collection setups, as variations in sensor placement, signal quality, and ground truth acquisition are likely across labs and real-world deployments (5). In practice, a more viable approach may involve using a pretrained foundation model and performing lightweight adaptation, such as fine-tuning only later layers, to quickly personalize the system without requiring extensive retraining time.

Conclusion

Overall, linear models show surprisingly strong performance in the setting of user personalization while achieving extremely low latency. They degrade by only $\sim 2^\circ$ in angular error relative to the emg2pose benchmark, and even outperform the small neural model, which degrades by $\sim 4^\circ$. However, these models produce significantly less smooth outputs. As a proof of concept, EMA smoothing was applied, reducing jerk by approximately 80% and producing usable predictions. This further highlighted a key limitation of linear models, as they tend to average across outputs rather than capture fine-grained articulation. This behavior is reflected in the lower variance of per-joint error ($\sim 24 \text{ deg}^2$) compared to emg2pose ($\sim 64 \text{ deg}^2$), which attempts more detailed finger-level control. Though this can likely be mitigated by changing the optimization task of the model to per finger errors.

This tradeoff can be acceptable depending on the intended prosthetic application. Many devices do not support independent control of individual fingers due to mechanical design (14), so this simplified and more stable output may still be appropriate while retaining continuous control rather than being confined discrete gesture switching.

Finally, classical ML models did not retain performance under generalization to unseen users, while not the focus of this study, still provides insight into robustness. Even in user-specific settings, EMG signals exhibit variability over time, which should be considered when selecting a modeling approach.

Data Availability

- Dataset: emg2pose dataset released by Meta Reality Labs
- Pretrained model checkpoints: emg2pose checkpoints, providing trained model weights used for initialization and inference
- Code: [GitHub repository](#) for this work, containing the full experimental setup and evaluation scripts

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